⚡ SSC CGL — Arithmetic Trick Decision Matrix

Recognition → Shortcut → Formula → Fastest route. Built from the repository's PYQ pattern banks (2021–2025).

Target post: ASO · Tier-I + Paper-I scope · Solve each type in under 30 seconds.

🟢 Fastest method 🔴 Common trap / mistake 🔷 Butterfly (cross-multiply) trick 🔴 High-repeat topic

How to read every row: Match the question to a **Situation** → confirm via the **Recognition** cue (the words that give it away) → apply the **Shortcut/Formula** → the green column is the single fastest route in the exam → dodge the red trap. The Butterfly and **Ratio** columns are the two speed-tools that recur across almost every topic.

1 · Profit & Loss 🔴 High Repeat · 2–4 Q/paper

Golden rule: fix **CP = 100** (or CP = ₹100 per unit) and everything becomes a percentage you can read off directly.

SITUATION	GIVEN	ASKED	RECOGNITION CUE	SHORTCUT / FORMULA	RATIO / B
Basic P%/L%	CP & SP	Profit or loss %	"bought … sold …"	`P% = (SP−CP)/CP ×100`	Set CP=10
Find CP from SP	SP & P% (or L%)	CP	"gains x%, find CP"	`CP = SP×100/(100+P%)`	SP×(100/(clean fract
Mark-up + discount	x% above CP, y% discount	Net profit %	"marks … above, gives … discount"	`x − y − xy/100`	MP=140, S CP 100
Same SP, ±x%	2 items, +x% & −x%	Overall %	"one at gain x%, other at loss x%"	`Always LOSS = x²/100 %`	—
Loss = SP of n articles	"loses SP of n on selling N"	Loss %	"loses = SP of 3 articles"	Let SP=1; CP(N)=N+n; `Loss%=n/(N+n)×100`	SP=1 each
Dishonest weight	False weight	Gain %	"uses faulty/short weight"	`Gain% = Error/(True−Error)×100`	Error over

SITUATION	GIVEN	ASKED	RECOGNITION CUE	SHORTCUT / FORMULA	RATIO / B
	sold at CP				

2 · Percentage 🔴 Feeds P&L, SI/CI, DI

Convert every % to a fraction (25%=1/4, 12½%=1/8, 16⅔%=1/6). The fraction method is 3× faster than long multiplication.

SITUATION	GIVEN	ASKED	RECOGNITION CUE	SHORTCUT / FORMULA	RATIO / F
Price ↑, cut use	Price rises x%	% cut in consumption	"expenditure unchanged"	`x/(100+x)×100`	Rise 1/4 → 1/(4+1)=1
A is x% more than B	A vs B relation	B is how much less	"by what % is B less than A"	`x/(100+x)×100`	+25% ↔ −
Election margin	Winner w%, wins by V	Total votes	"won by … votes"	Margin=(2w−100)% ; `Total=V×100/margin`	V ÷ margi
↑then↓ same x%	+x% then −x%	Net change	"increased then decreased"	`net = −x²/100 %` (always loss)	Factor 1.1
Successive %	a% then b%	Single equivalent	two chained changes	`a + b + ab/100`	Multiply d
Population back-track	+p%, −q% over years	Original	"present population is …"	P × factor_1 × factor_2 = now	Divide nov factors

3 · Simple & Compound Interest 🔴 High Repeat

For 2 years, the whole chapter reduces to two memorized results. For rate-finding, read the amount ratio as a perfect square/cube.

SITUATION	GIVEN	ASKED	RECOGNITION CUE	SHORTCUT / FORMULA	RATIO TRI
CI−SI, 2 yr	Difference, r, t=2	Principal	"difference between CI and SI, 2 years"	`Diff = P(r/100)²`	Diff = P × $(r\%)^2$ direct
CI−SI, 3 yr	Difference, r, t=3	Principal	"…for 3 years"	`Diff = P(r/100)²·(3+r/100)`	—
Find rate (CI)	P → A in 2 yr	Rate	"amounts to … in 2 years CI"	`A/P = (1+r/100)²`	Read A/P a a square
Amount at two times	A at t and at 2t	Principal	"becomes X after 3 yr, Y after 6 yr"	Ratio Y/X = $(1+r)^t$; P=X/(that)	Y/X gives th growth factor for t
Doubling / n-fold	Sum doubles in t yr (CI)	Time to 4×, 8×	"doubles … becomes 4 times"	2× in t → 4× in 2t → 8× in 3t	Powers of the doublir time

4 · Time & Work + Pipes & Cisterns 🔴 Every paper

LCM method beats fractions every time. Total Work = LCM(days). Efficiency = Total ÷ days → whole numbers → add/subtract freely.

SITUATION	GIVEN	ASKED	RECOGNITION CUE	SHORTCUT / FORMULA	LCM / RA
Two together	A=a days, B=b days	Combined time	"working together"	`ab/(a+b)`	LCM: add efficiencie Total÷sun
Find one alone	(A+B) & B	A alone	"together …, B alone …"	`1/A = 1/T − 1/B`	LCM: eff^A eff^B
Pairwise (A+B,B+C,C+A)	3 pair-times	All-three time	"A&B in…, B&C in…, C&A in…"	Sum pairs = 2(A+B+C)	LCM: pair the sum
% more efficient	A is x% more than B	Other's time	"25% more efficient"	eff 125:100=5:4 → time 4:5	Time ratic (100+x)

SITUATION	GIVEN	ASKED	RECOGNITION CUE	SHORTCUT / FORMULA	LCM / RA
Leave / join midway	Some work, then change	Remaining days	"after k days, X leaves/joins"	Units done → Total − done → ÷ new rate	Track unit LCM boar
Men–Days–Hours	$M_1D_1H_1$, work scaled	New M/D/H	"×larger work, hrs/day"	$M_1D_1H_1/W_1 = M_2D_2H_2/W_2$	Keep mar constant
Pipe + leak	Fill x, empty y	Net fill time	"leak empties in …"	$Net = 1/x - 1/y$ (inlet +, leak −)	LCM: +eff −eff for le
Leak from delay	Fills in x, took x′ with leak	Leak-empty time	"due to leak took longer"	$Leak = x \cdot x'/(x' - x)$	1/x − 1/x′ rate

5 · Partnership

Profit share ∝ **capital × time**. Reduce to a clean ratio, then split. Deduct any salary/commission *before* ratio-splitting.

SITUATION	GIVEN	ASKED	RECOGNITION CUE	SHORTCUT / FORMULA	RATIO / BUTT
Equal time	Capitals, same period	Profit split	"invested for a year each"	Share ∝ capital	Ratio of capita
Different times	Capital & months differ	Profit split	"A for 12 mo, B for 8 mo"	$Share \propto C \times T$	Multiply C×T, t ratio
Work-done share (link)	Wages for a job	Each share	"earn ₹… for the work"	Wages ∝ work = eff×days	C does 4/11 → of money
Working partner	Salary + capital ratio	Net split	"gets salary for managing"	Deduct salary → split remainder by C×T	Remaining ÷ equivalent cap

6 · Mixture & Alligation

Alligation is the fastest tool whenever a **mean/average value** sits between two ingredient values — it gives the mix ratio in one cross.

SITUATION	GIVEN	ASKED	RECOGNITION CUE	SHORTCUT / FORMULA	ALLIGATION
Two-ingredient mix	Cheap c, dear d, mean m	Mix ratio	"mixed to cost/average …"	`cheap:dear = (d−m):(m−c)`	Cross-subtrac
Milk–water dilution	Add water free	Ratio / profit	"adds water and sells at CP"	Water is 0-cost ingredient in cross	Treat water va
Repeated replacement	Remove & refill n times	Final pure qty	"drawn off and replaced, n times"	`Final = V(1−r/V)ⁿ`	Ratio (1−r/V)
Two mixtures combined	Ratios in each vessel	Final ratio	"mix vessel A and B"	Find each part's fraction, alligate on the fraction	Cross on the r

7 · Speed · Time · Distance 🔴 High Repeat

One conversion unlocks most sums: `km/h → m/s ×5/18` · `m/s → km/h ×18/5`.

SITUATION	GIVEN	ASKED	RECOGNITION CUE	SHORTCUT / FORMULA	RATIO TRI
Equal-distance avg	Speeds x, y (same dist)	Average speed	"half journey at …, half at …"	`2xy/(x+y)` (harmonic)	Never the plain mean
Late / early	Two speeds, ±minutes	Distance	"late at …, early at …"	Time gap = (late+early); d/x − d/y = gap	Add late+early total gap
Same-time meeting	Two bodies start together	Distances / total	"meet after starting together"	Distance ∝ speed (same time)	$d_1:d_2 = s_1:s$

SITUATION	GIVEN	ASKED	RECOGNITION CUE	SHORTCUT / FORMULA	RATIO TRI
Ratio of speeds ↔ time	Fixed distance	Time ratio	"same distance, speeds a:b"	Time ∝ 1/speed	speeds a:b times b:a

8 · Trains

Decide the **distance to cover** first: pole = train length only; platform/bridge = length + platform; passing another train = **sum** of both lengths.

SITUATION	GIVEN	ASKED	RECOGNITION CUE	SHORTCUT / FORMULA	RATIO TRIC
Cross a pole/man	L, t	Speed	"crosses a pole/standing man"	`S = L/t`	—
Cross platform/bridge	L, P, t	Speed / length	"crosses a platform/bridge"	`S = (L+P)/t`	—
Pole then platform	t_1 (pole), t_2 (platform P)	Length	"pole in …, platform in …"	Extra time (t_2-t_1) covers P → $S=P/(t_2-t_1)$	Platform ÷ extra time = speed
Two trains, opposite	L_1,L_2, speeds	Crossing time	"opposite directions"	Rel speed = S_1+S_2; dist = L_1+L_2	Add speeds, add lengths
Two trains, same dir	L_1,L_2, speeds	Overtake time	"same direction / overtakes"	Rel speed = S_1-S_2; dist = L_1+L_2	Subtract speeds, add lengths

9 · Boats & Streams

Two back-solve identities crack almost everything: `b = (down+up)/2` and `s = (down−up)/2`.

SITUATION	GIVEN	ASKED	RECOGNITION CUE	SHORTCUT / FORMULA	RATIO TR
Basic up/down	b (still), s (stream)	Down/up speed	"downstream / upstream"	`down=b+s, up=b−s`	—

SITUATION	GIVEN	ASKED	RECOGNITION CUE	SHORTCUT / FORMULA	RATIO TR
Back-solve b & s	down & up speeds/times	b or s	"...km down in ...h, back in ...h"	`b=(d+u)/2, s=(d−u)/2`	Convert e leg to spe first
Up = k × down time	Time ratio, s known	b	"twice/thrice as long upstream"	down speed = k × up speed → solve	(b+s)=k(b
There-and-back time	b, s, total time	Distance	"to a place and back in … h"	d/(b+s) + d/(b−s) = T	Sum the t leg-times

10 · Ratio & Proportion

SITUATION	GIVEN	ASKED	RECOGNITION CUE	SHORTCUT / FORMULA	BUTTI
Combine A:B, B:C	Two chained ratios	A:B:C	"A:B=..., B:C=..."	Make B common (cross-multiply)	×B-ter
Divide by ratio	Total, parts	One share	"divide ₹... in ratio ..."	Each part = Total ÷ Σparts	Multip part-v
Proportionals	a,b (,c)	Mean/3rd/4th	"mean/third/fourth proportional"	mean √(ab); 3rd b^2/a; 4th bc/a	ad=bc
Componendo-dividendo	a/b = c/d	(a+b)/(a−b)	fraction with ± on both	`(a+b)/(a−b)=(c+d)/(c−d)`	Add/s & den

11 · Average

SITUATION	GIVEN	ASKED	RECOGNITION CUE	SHORTCUT / FORMULA	RATIO / SHC
Add one value	avg of n, new value	New average	"one more is included"	Δavg = (new − old avg)/(n+1)	Shift, don't re

SITUATION	GIVEN	ASKED	RECOGNITION CUE	SHORTCUT / FORMULA	RATIO / SHC
Consecutive numbers	First & last	Average	"consecutive / evenly spaced"	(first+last)/2	Middle term average
Replace a member	n, old, new person	Age/weight change	"replaced by, average changes"	New−Old = n × Δavg	Total shift sp over n

★ Master Recognition Table — "Which trick, how fast?"

Skim this the night before the exam. Left = the words in the question; right = the reflex tool.

IF THE QUESTION SAYS…	INSTANT TOOL
"expenditure unchanged / consumption"	Fraction: rise 1/n → cut 1/(n+1)
"gain x%, find CP"	SP×100/(100+x) as a reduced fraction
"marks above & gives discount"	x − y − xy/100
"CI − SI for 2 years"	$P(r/100)^2$
"amounts to … in 2 yr CI"	Read A/P as a perfect square
"working together"	LCM board / $ab\cdot(a+b)^{-1}$
"x% more efficient"	Flip eff-ratio → time-ratio
"leak / pipe empties"	Signed LCM (+fill, −leak)
"invested for different months"	Equivalent capital = C×T
"mixed to average / mean value"	Alligation cross
"drawn off and replaced n times"	$V(1-r/V)^n$
"half at …, half at …" (speed)	2xy/(x+y)
"late … early …"	gap = late+early, one equation
"crosses pole vs platform"	Extra time × speed = platform
"downstream … upstream …"	b=(d+u)/2, s=(d−u)/2

IF THE QUESTION SAYS...	INSTANT TOOL
"start together & meet"	distance ∝ speed
"A:B and B:C"	Make B common (cross-multiply)

Three universal speed-tools that reappear in every topic above:
◆ Butterfly / cross-multiplication — combining ratios, alligation, C×T partnership, proportion (ad=bc).
▣ **LCM board** — Time&Work, Pipes; turn every rate into a whole number.
⚖ **Ratio inversion** — efficiency↔time, speed↔time; whatever is "more" of one is "less" of the other in exact inverse.

⚠ Multi-step "Chaos" Problems — where two or three topics collide

Why these are hard: SSC stacks 2–3 concepts in one long sentence so you lose the thread. **Fix:** don't solve the sentence — **break it into numbered steps**, finish each with a clean number, then chain them. The green line is the fast route; the red line is where most candidates lose the mark.

LONG PROBLEM (READ ONCE, THEN CHUNK IT)	TOPICS STACKED	STEP-BY-STEP
A shopkeeper marks his goods 40% above cost, then allows a 25% discount on the marked price, and finally, because the customer pays cash, gives a further 4% discount on that reduced price. If his cost was ₹1,000, what is his overall profit or loss percent?	P&L successive %	① MP = 1000×1.4(1400. ② After 25%: 1400 = 1050. ③ After 4% more: 1050×0.96 = 1008. ④ SP 1008 on CP 1
A and B can finish a piece of work in 12 days; B and C in 15 days; C and A in 20 days. All three start together, but after 4 days A leaves, and 3 days later B also leaves, C finishing the rest alone. In how many total days is the work completed?	Time & Work pairwise leave-midway	① LCM(12,15,20)=(Pairs: A+B=5, B+C= C+A=3 → 2(A+B+(→ all=6; A=2,B=3,C ② First 4 days all tl 6×4=24 units. ③ Next 3 days B+C left): 4×3=12 → tot ④ Left 60−36=24 l alone: 24/1=24 day

LONG PROBLEM (READ ONCE, THEN CHUNK IT)	TOPICS STACKED	STEP-BY-STEP
A train 180 m long, running at 72 km/h, first crosses a man walking at 9 km/h in the opposite direction, and then, a minute later, crosses a 270 m platform. What is the ratio of the time it takes in the first crossing to the second?	Trains relative speed ratio	① vs man (opposit = 72+9 = 81 km/h 22.5 m/s; dist = 18 → t_1 = 180/22.5 = 8 ② vs platform: spe 20 m/s; dist = 180- = 450 → t_2 = 450/2 22.5 s. ③ ratio = 8 : 22.5 = 45.
The population of a town was 30,000 two years ago. It rose by 10% in the first year, fell by 10% in the second, and this year the municipality expects a further rise of 20%. If 8% of the final population will be senior citizens, roughly how many senior citizens are expected?	% successive of-value	① ×1.10×0.90 = × → 30000×0.99 = 29 ② ×1.20 → 29,700 = 35,640. ③ 8% of 35,640 = 2,851.2 ≈ 2,851.
A sum of money is divided among A, B and C so that A gets ₹1,800, which is 3/13 of the whole. The remaining amount is invested at compound interest at 10% per annum. What is the compound interest earned on the remainder after 2 years?	Ratio CI	① A = 3/13 of tota 1800 → total = 1800×13/3 = 7,800 ② Remainder = 7,8 1,800 = 6,000. ③ CI 2 yr @10%: 6000×(1.21−1) = 6000×0.21 = 1,260
A boat goes 24 km downstream in 3 hours and returns the same distance upstream in 4 hours. If the boat now has to travel 60 km downstream, and the stream speed suddenly doubles for the second half of that trip, how long does the 60 km downstream leg take?	Boats back-solve 2-phase	① down = 24/3 = = 24/4 = 6 → b = 7 1. ② First 30 km at d = b+s = 8 → 30/8 3.75 h. ③ Stream doubles (s=2): down = 7+2 → 30/9 = 3.33 h. ④ Total = 3.75 + 3 7.083 h.

The chunking habit: circle each number as you read, write ① ② ③ down the margin, and never carry a fraction forward when the LCM board or a clean ₹/km value is available. One clean number per step = no chaos.